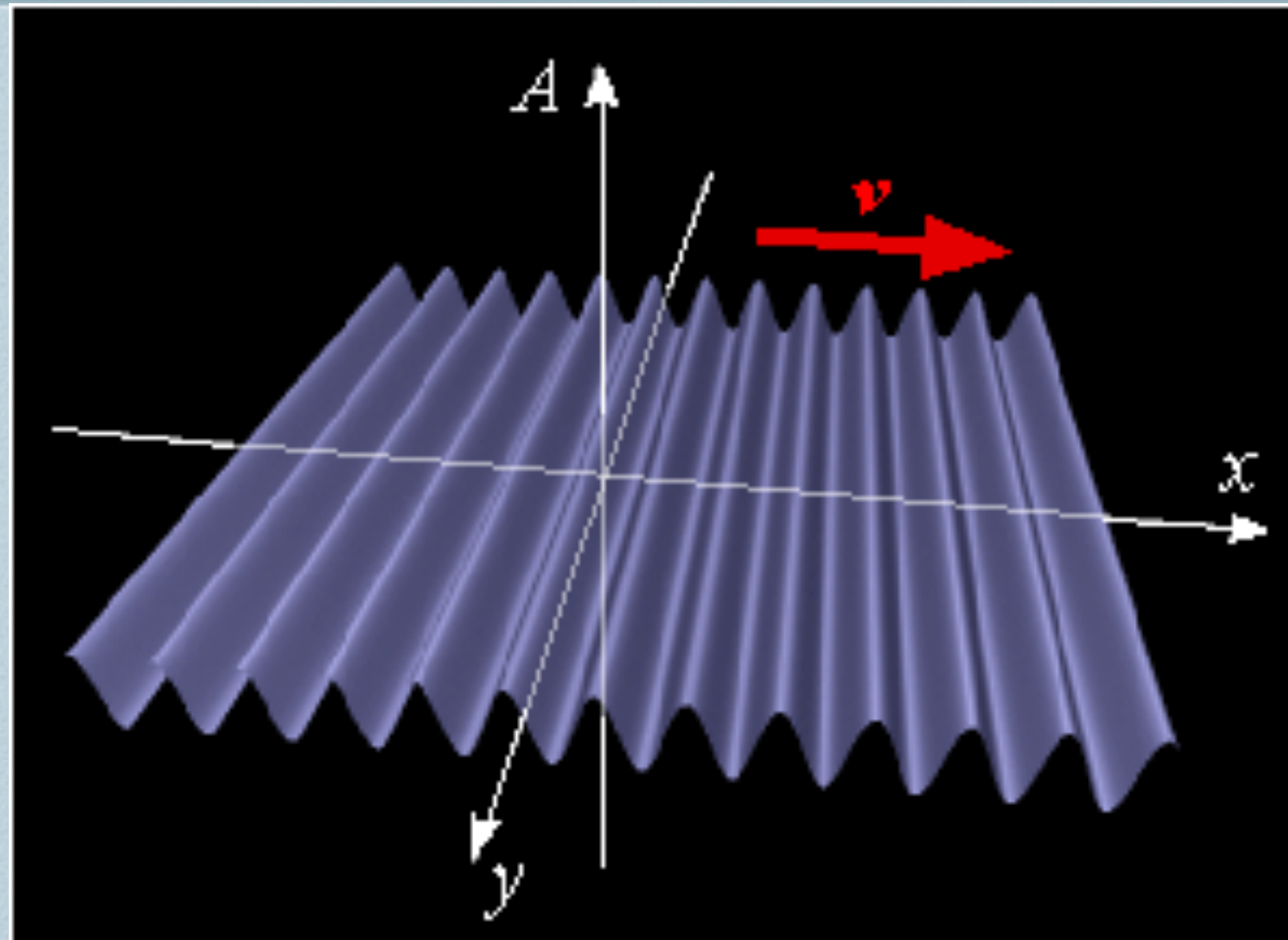


**CREATE Module:
Perturbation Theory
Lecture 8: WKB theory via the shallow water
equations**

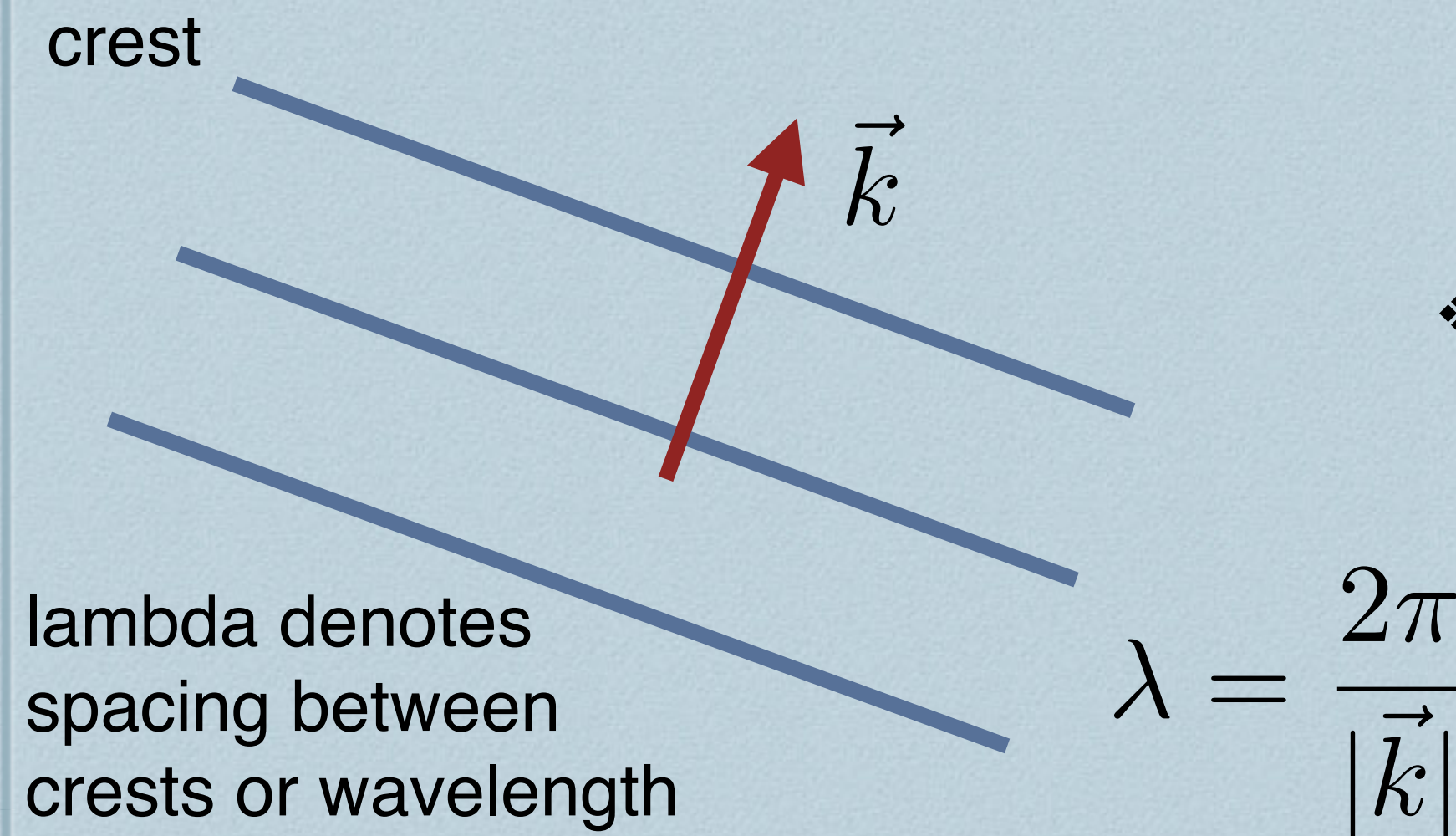
Marek Stastna

- ❖ WKB theory is a standard topic in “traditional” perturbation theory courses.
- ❖ However, it is generally presented for an ODE and this is not really representative of its use.
- ❖ In these notes I develop some basics of wave theory from physics and then use a classical set of PDEs, the linearized shallow water equations, to discuss how WKB theory comes about as an answer to a concrete question.
- ❖ I conclude with a derivation of the so-called ray tracing equations.
- ❖ But I begin by discussing the physical description of waves.
- ❖ In the typically mathematics undergraduate curriculum we are taught that waves are governed by the “wave equation” which is a second order, hyperbolic equation.
- ❖ In reality only a subset of real world waves are “hyperbolic”.

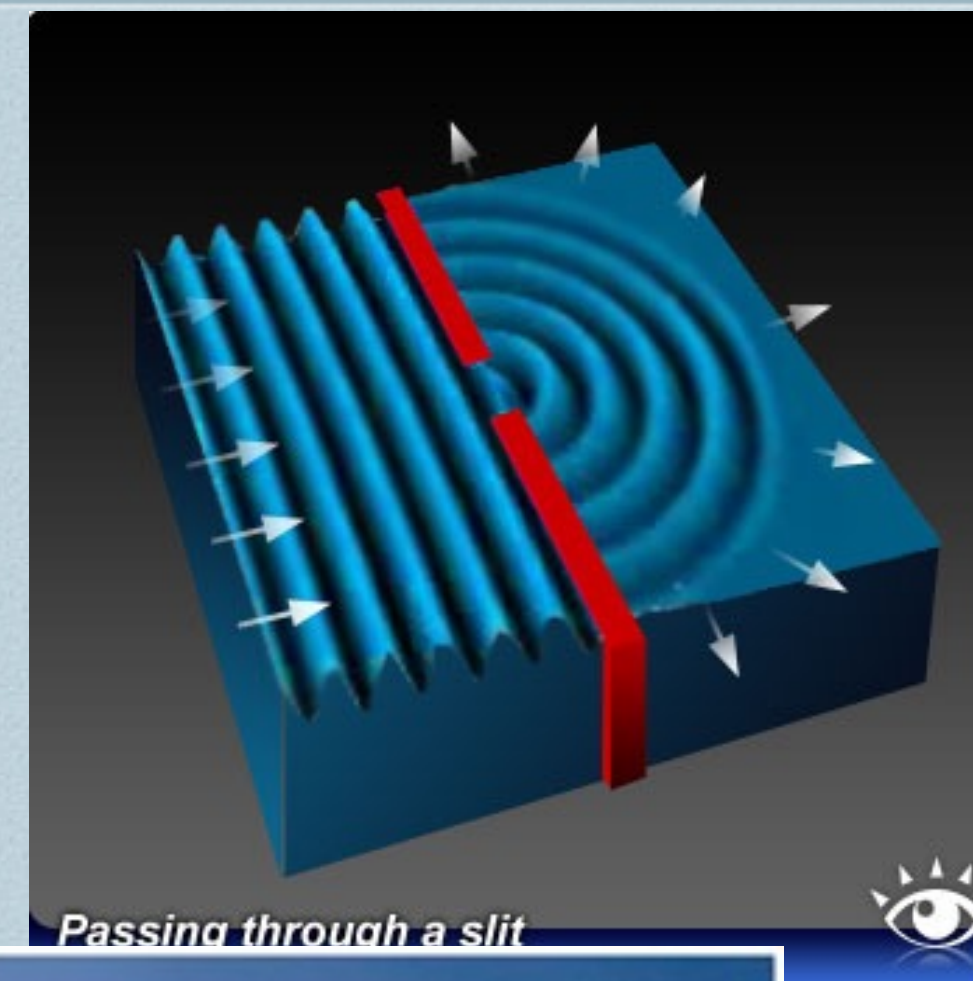
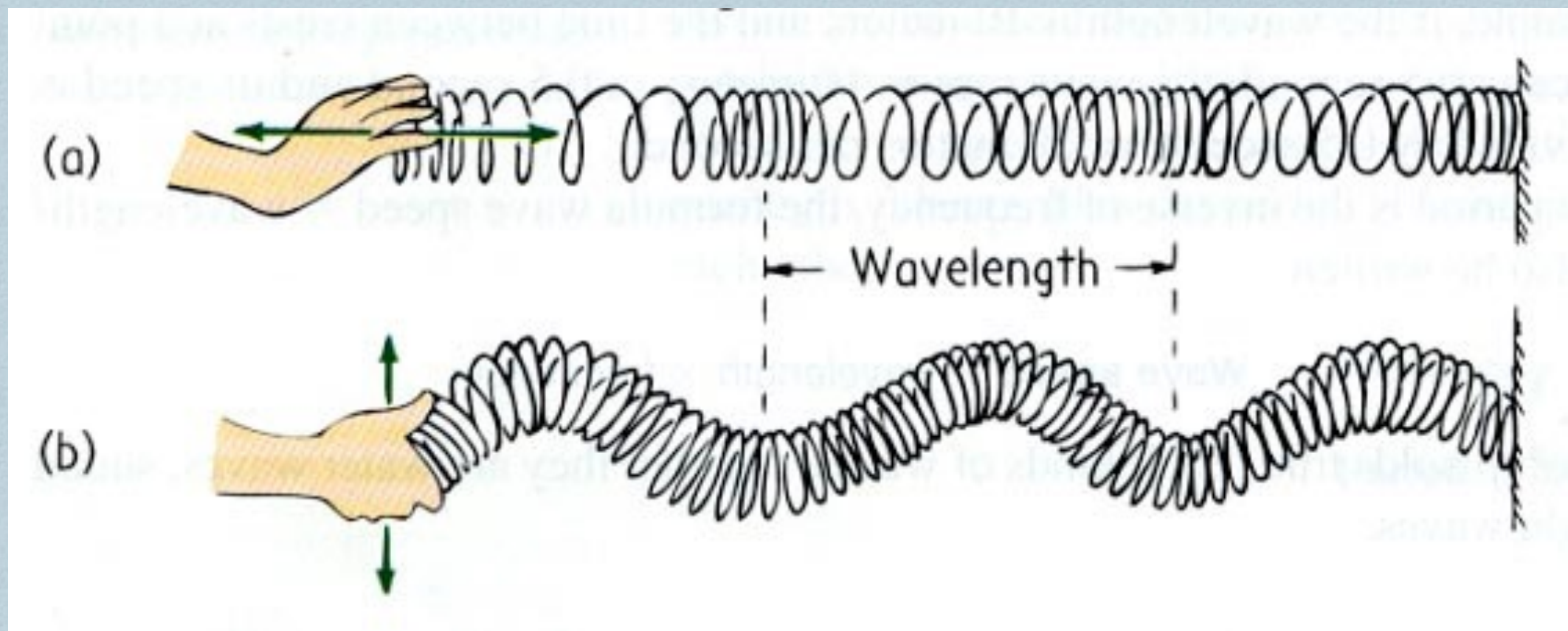


$$A(x, t) = \text{Re}\{\exp[i(\vec{k} \cdot \vec{x} - \sigma t)]\}$$

- ❖ Plane waves are a basic building block of physical theories (next slide has examples).
- ❖ The direction of propagation and the inter-crest spacing is given by the wavenumber vector.
- ❖ In general the frequency may depend on wavenumber, though often the first examples we see (e.g. EM waves in a vacuum) are so-called non-dispersive waves.



- ❖ Mathematicians are often presented “waves” as solutions to the wave equation which is a bit unphysical.
- ❖ The wave equation is an example of a hyperbolic equation and such equations should really be thought of as representing signals (the information carried along characteristics), as opposed to waves.

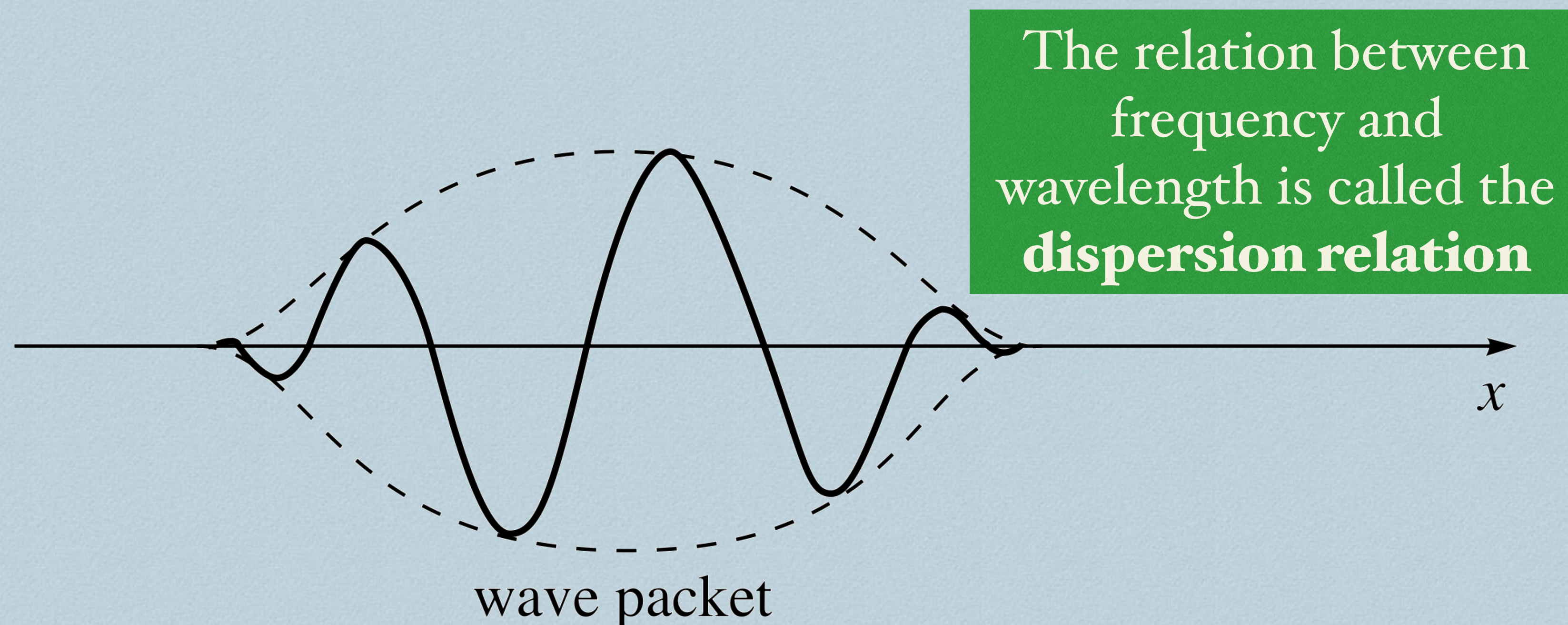


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Various
wave
phenomena:
For interest
only

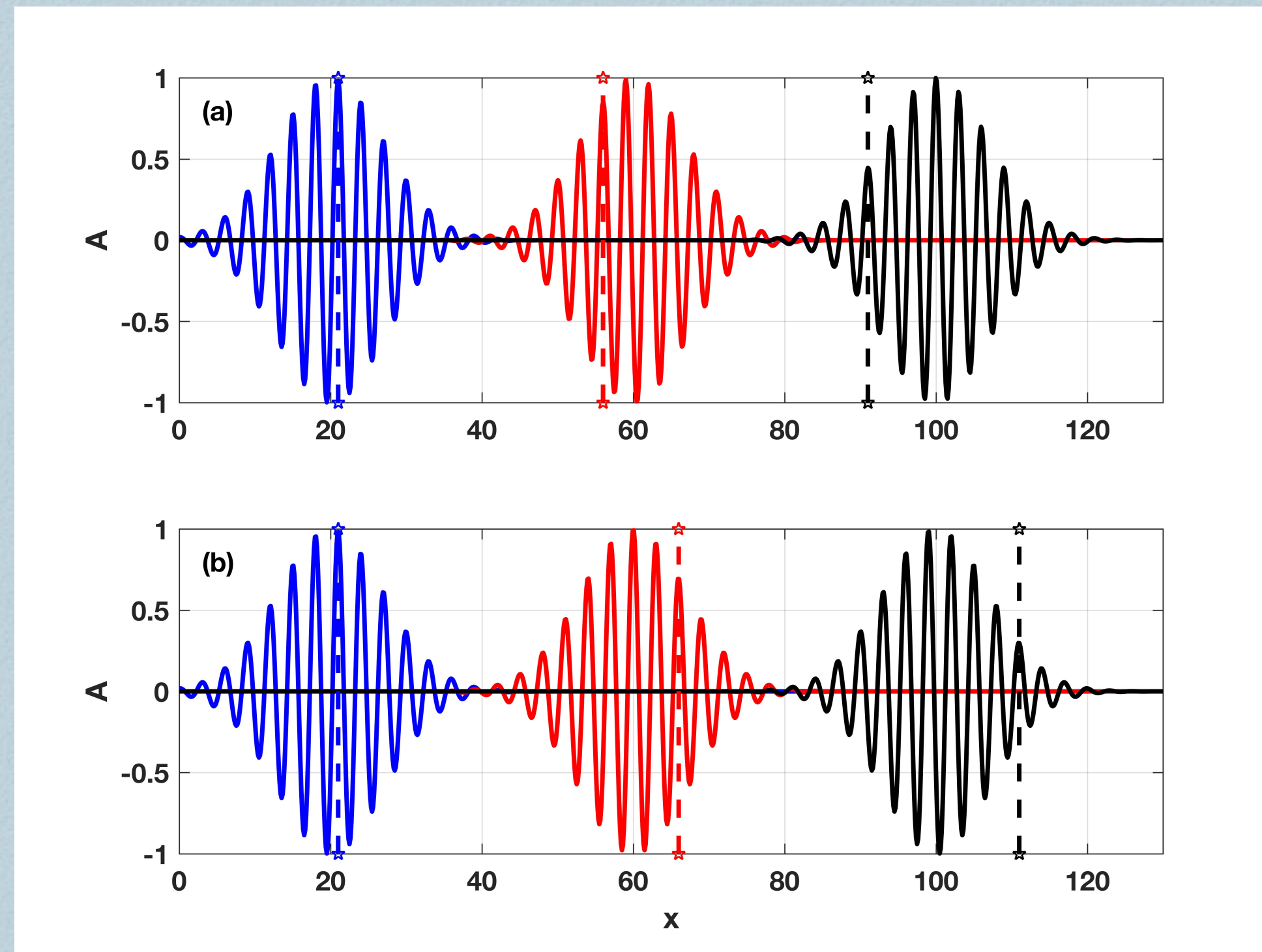
- ❖ Plane waves are quite idealized since they have no beginning or end, and often we are concerned with localized disturbances.
- ❖ These too have a well-known representation in physics called a **wave packet**.
- ❖ A wave packet consists of a plane wave and an envelope. The envelope (dashed line below) is assumed to be much longer, and much more slowly varying than the plane wave (solid line below).

$$A(\epsilon x, t) \cos(kx - \sigma(k)t) \text{ where } \epsilon \ll 1$$



- ❖ When frequency depends on wavenumber there are two velocities (or speeds in 1D).
- ❖ The velocity of the wave crest, called the **phase velocity**, which is largely a mathematical construct.
- ❖ The velocity of the wave packet, or of energy propagation, called the **group velocity** which is the more physical concept since energy moves with the **group velocity**.

$$c_p < c_g$$



$$c_p > c_g$$

$$c_p = \frac{\sigma}{k}$$

The dashed vertical line moves with the phase velocity

$$c_g = \frac{d\sigma}{dk}$$

The packet moves with the group velocity

Examples of dispersion relations are shown on the next slide

$$\sigma = \pm \sqrt{gk} \quad c_p = \pm \sqrt{\frac{g}{k}} \quad c_g = \pm \frac{1}{2} \sqrt{\frac{g}{k}}$$

- ❖ Long waves travel faster, energy propagates slower than crests.
- ❖ Fairly common in nature, as we will see for deep water waves.

Bottom row
on last
slide

$$\sigma = \pm \alpha k^{3/2} \quad c_p = \pm \alpha \sqrt{k} \quad c_g = \pm \alpha \frac{3}{2} \sqrt{k}$$

- ❖ Short waves travel faster, energy propagates faster than crests.
- ❖ Not common apart from surface tension (or capillary waves).

Top row
on last
slide

- ❖ Dispersion modified advection equation (a “model” equation)

$$A_t = -cA_x - \beta A_{xxx}$$

$$A = \exp(i[kx - \sigma(k)t])$$

$$-i\sigma(k) = -cik + \beta ik^3$$

$$\sigma(k) = ck - \beta k^3$$

$$c_p = c - \beta k^2$$

$$c_g = c - 3\beta k^2 \leq c_p$$

- ❖ In environmental and geophysical fluid mechanics, the shallow water equations are a very famous system of PDEs.
- ❖ This is because they are analogous to the equations of gas dynamics that (unlike gas dynamics) can be derived without reference to thermodynamics, and thus provide a convenient system for studying shocks.
- ❖ Shocks are nonlinear waves, and for our purposes we will consider only small amplitude waves for which it is reasonable to linearize.
- ❖ The 1 dimensional, linearized shallow water equations in a fluid of variable depth $H(x)$ for the free surface height, η , and velocity u read:

Conservation of
momentum

$$\frac{\partial u}{\partial t} = -g \frac{\partial \eta}{\partial x}$$

Conservation of
mass

$$\frac{\partial \eta}{\partial t} + \frac{\partial u H(x)}{\partial x} = 0$$

$$\frac{\partial^2 \eta}{\partial t^2} = \frac{\partial}{\partial x} \left(g H(x) \frac{\partial \eta}{\partial x} \right)$$

$$\frac{\partial^2 \eta}{\partial t^2} = \frac{\partial}{\partial x} \left(c^2(\epsilon x) \frac{\partial \eta}{\partial x} \right) \quad \epsilon \ll 1 \quad c(\epsilon x) = \sqrt{gH(\epsilon x)}$$

- ❖ Next assume that the total depth varies slowly and look for solutions motivated by the linear dispersive theory.
- ❖ The linear dispersive theory would look for solutions of the form:
 $a_0 \exp(i(kx - \sigma(k)t)) = a_0 \exp(i\Theta)$
- ❖ The second form gives us the relations $k = \frac{\partial \Theta}{\partial x}, \sigma = -\frac{\partial \Theta}{\partial t}$
- ❖ The perturbation method now seeks solutions that generalize the above as:
 $\eta = a(\epsilon x) \exp(i\Theta(x, t, \epsilon x))$
- ❖ In these notes I am first going to follow the WKB formalism, even though the problem I posed here has solutions that go back to the mid 1800s (The Liouville-Green theory).

$$\eta_{tt} = g \left(H(x) \eta_x \right)_x \implies \eta_{tt} = gH(x) \eta_{xx} + gH'(x) \eta_x$$

- ❖ WKB starts with the general PDE with a reference depth H_0 so that $c_p^2 = \sigma_0^2/k^2 = gH_0$

- ❖ It then separates variables as $\eta(x, t) = \hat{\eta}(x) \exp(-i\sigma_0 t)$ to get

$$\frac{d^2 \hat{\eta}}{dx^2} + \frac{1}{H(x)} \frac{dH}{dx} \frac{d\hat{\eta}}{dx} + \frac{\sigma_0^2 \hat{\eta}}{gH(x)} = 0$$

- ❖ We then non-dimensionalize via $\hat{\eta} = A\tilde{\eta}$, $x = L\tilde{x}$, $H = H_0\tilde{h}$ and note that by linearity A will cancel

$$\frac{1}{L^2} \left(\frac{d^2 \tilde{\eta}}{d\tilde{x}^2} + \frac{1}{\tilde{h}} \frac{d\tilde{h}}{d\tilde{x}} \frac{d\tilde{\eta}}{d\tilde{x}} \right) + k^2 \frac{\tilde{\eta}}{\tilde{h}} = 0$$

- ❖ Drop tildes and define $\epsilon = 1/(kL)$. This is effectively the assumption of slow spatial variation in $H(x)$ and hence the wave train. This leaves us the final ODE:

$$\epsilon^2 \left(\eta''(x) + \frac{h'}{h} \eta' \right) + \frac{\eta}{h} = 0$$

ODE from last slide $\epsilon^2 \left(\eta''(x) + \frac{h'}{h} \eta' \right) + \frac{\eta}{h} = 0$

❖ This is reminiscent of the singular problem for polynomials, so we expect a $\frac{1}{\epsilon}$ to appear somewhere.

❖ We seek solutions of the form $\eta(x) = \exp(S(x))$ where $S(x)$ has an asymptotic expansion

$$S(x) \sim \frac{S_0(x)}{\epsilon} + S_1(x) + \epsilon S_2(x)$$

❖ Substitution gives an equation in $S(x)$:

$$\epsilon^2 \left(S''(x) + (S'(x))^2 + \frac{h'}{h} S'(x) \right) + \frac{1}{h} = 0$$

❖ And the rest is just lots of algebra. At leading order we get:

$$\left(\frac{dS_0}{dx} \right)^2 = -\frac{1}{h} \implies S_0(x) = \pm i \int_0^x \frac{1}{\sqrt{h(y)}} dy$$

- ❖ At first order we get:

$$S_0''(x) + 2S_0'(x)S_1'(x) + (h'/h)S_0'(x) = 0$$

Note you can solve
for S_1 in terms of
 S_0

- ❖ And a bit of algebra shows (C is a constant of integration):

$$S_1(x) = \ln h^{-1/4}(x) + C$$

- ❖ The $\ln$ term will cancel with the exponential giving us the change in from the want. Substitution into $\eta(x) = \exp(S_0(x)/\epsilon + S_1(x))$ gives

$$\eta(x) = \frac{a_{\pm}}{h^{1/4}} \exp \left(\pm \frac{i}{\epsilon} \int_0^x \frac{1}{\sqrt{h(y)}} dy \right)$$

- ❖ And finally we can substitute back and get the dimensional expression:

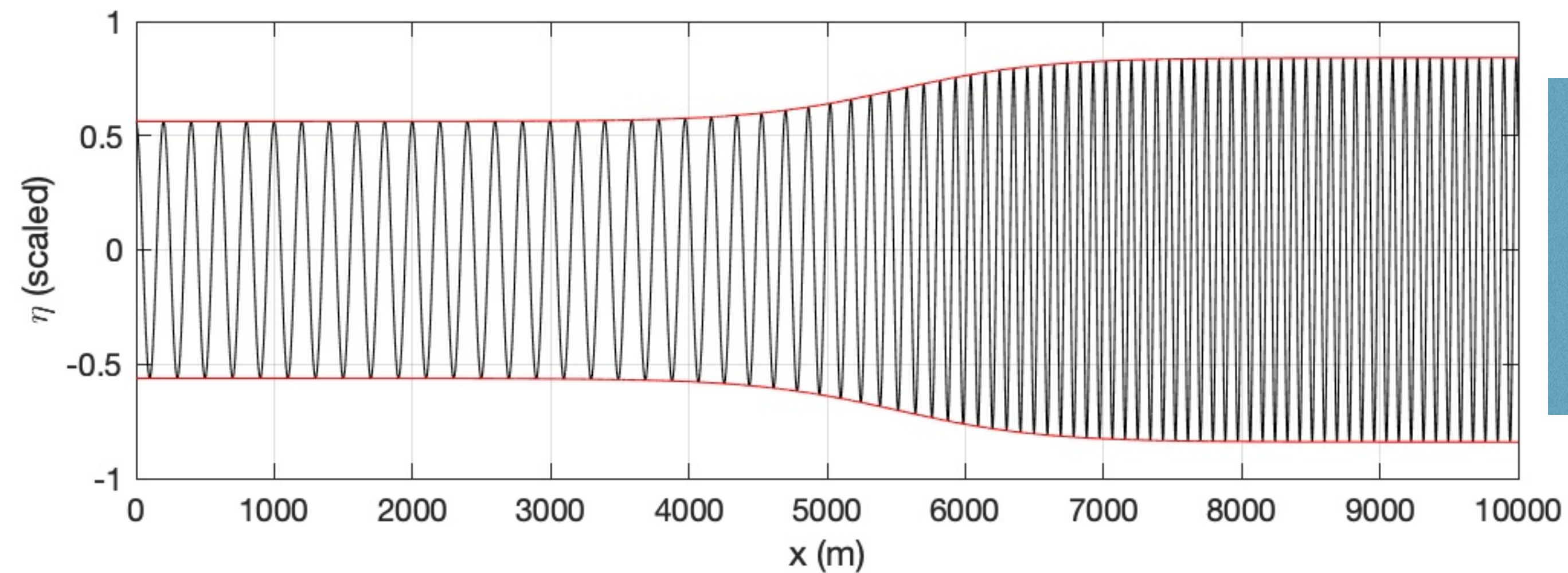
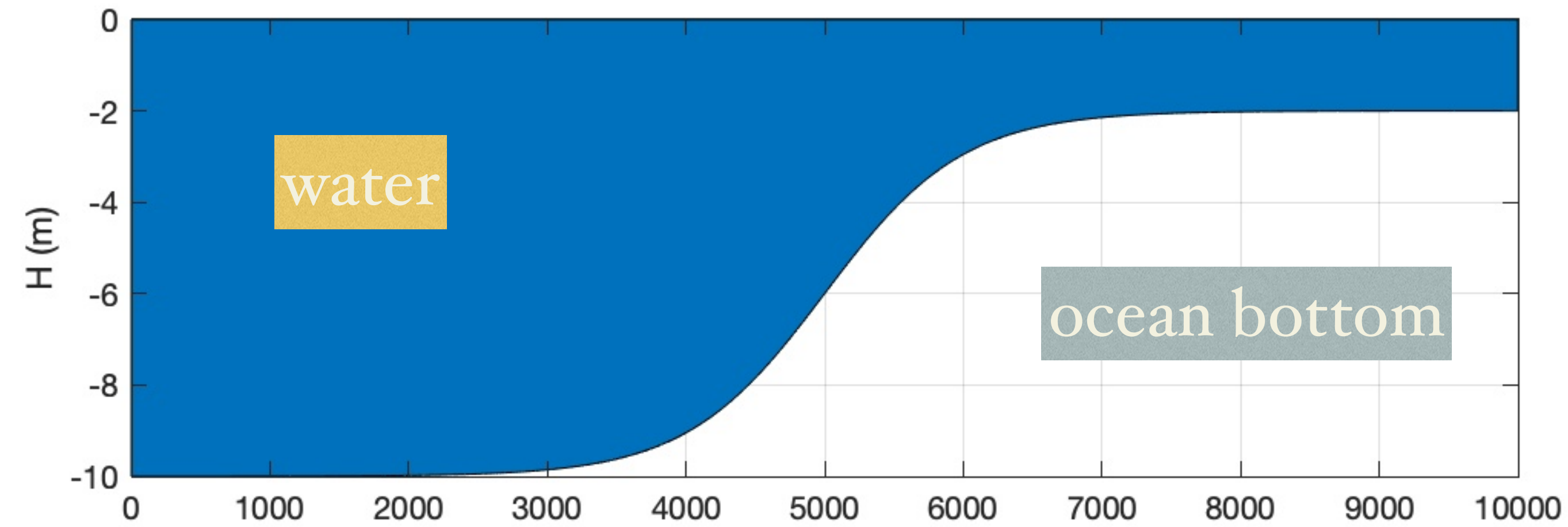
Amplitude

Like “kx”

$$\eta(x, t) = \frac{C_{\pm}}{H^{1/4}} \exp \left(\pm i k_0 \int_0^x \sqrt{\frac{H_0}{H(y)}} dy - i \omega_0 t \right)$$

- ❖ This is a solution of the form $\eta = a(\epsilon x) \exp(i\Theta(x, t, \epsilon x))$ in which the amplitude and phase are slowly varying.
- ❖ The expression for amplitude is sometimes known as Green's law.
- ❖ The local wavenumber is (by chain rule): $k = \frac{\partial \Theta}{\partial x} = k_0 \sqrt{\frac{H_0}{H(x)}}$
- ❖ Physicists define the wave energy as proportional to amplitude squared, and Green's law then gives that wave energy is proportional to one over the square root of the total depth $H(x)$.
- ❖ WKB theory can give more (e.g. for tunneling), but this is a good introduction.

Sample WKB solution: $H_0=10$ m, shallowest $H=2$ m.
Initial wave length 200m



As water gets shallower
waves grow in
amplitude and get
shorter

Shallow water equations without WKB: Ray Theory

Marek Stastna

- ❖ WKB theory is pretty formal, so are there other ways physicists have thought about slowly changing waves?
- ❖ Indeed there are. The most intuitive to me is the notion of slowly changing waves.
- ❖ For plane dispersive waves the wave ansatz in 2D is $a \exp(i(kx + ly - \sigma t))$.
- ❖ If we want the environmental conditions to change slowly in space or time we could write: $A(x, t)\exp(i\theta(x, t))$
- ❖ And using this define: $k = \partial_x \theta$, $\sigma = -\partial_t \theta$.
- ❖ You can think of these as the local wavenumber and local frequency.
- ❖ Because of the equality of mixed partials we have: $\partial_t k + \partial_x \sigma = 0$
- ❖ But we also have the dispersion relation: $\sigma = \Omega(k(x, t))$ so that: $\partial_t k + \partial_k \Omega \partial_x k = 0$
- ❖ Using the definition of group speed this gives: $(\partial_t + c_g \partial_x)k = 0$ or the fact that the wave number is advected by the group speed

- ❖ On the last slide we ended with: $(\partial_t + c_g \partial_x)k = 0$ or the fact that the wave number is advected by the group speed.
- ❖ Now let's say the frequency depends on a slowly varying parameter, like the depth in the WKB example: $\sigma = \Omega(k(x, t), H(x, t))$.
- ❖ Applying the Chain rule gives: $(\partial_t + c_g \partial_x)k = -\partial_H \Omega \partial_x H$
- ❖ And a similar calculation (exercise) gives: $(\partial_t + c_g \partial_x)\sigma = \partial_H \Omega \partial_t H$
- ❖ These are called the ray equations. They tell you that the ray itself is determined as the tangent curve to the group velocity (kind of trivial in 1D), and an observer moving with the ray would see changes in k due to spatial variations in H , and changes in σ due to temporal variations in H .
- ❖ So for the WKB depth change problem, with $H(x)$, we have that the frequency will not change along the ray, but the wave number will.

- ❖ Physicists usually add one more equation, derived from the linearized governing equations (things like the Conservation of Momentum).
- ❖ This is the conservation of wave energy, $E = \frac{1}{2}\sigma^2 a^2$, and reads: $\partial_t E + \partial_x(c_g E) = 0$ so that energy is conserved following a ray. For the variable depth shallow water problem we would have the equations:

$$\sigma = \sqrt{gH(x)}k \quad c_g = \frac{d\sigma}{dk} \quad \text{Dispersion relation and group speed}$$

$$\frac{dx}{dt} = c_g$$

Ray Path

$$(\partial_t + c_g \partial_x)k = -\frac{1}{2}\sqrt{\frac{g}{H}}kH'(x) \quad \text{Change of k along the ray}$$

$$(\partial_t + c_g \partial_x)\sigma = 0 \quad \text{Change of frequency along the ray}$$

$$\partial_t E + \partial_x(c_g E) = 0 \quad \text{Change of amplitude along the ray}$$

- ❖ The best part of these equations is that they let you solve step by step.
- ❖ As such they make it way to generalize to more complicated situations (next slide).

$$\sigma = \sqrt{gH(x)}k$$

$$c_g = \sqrt{gH(x)}$$

$$\frac{dx}{dt} = \sqrt{gH(x)}$$

$$(\partial_t + c_g \partial_x)k = \frac{d}{dx} \sqrt{gH(x)}k$$

$$(\partial_t + c_g \partial_x)\sigma = 0$$

$$\partial_t E + \partial_x (c_g E) = 0$$

Dispersion relation and
group speed

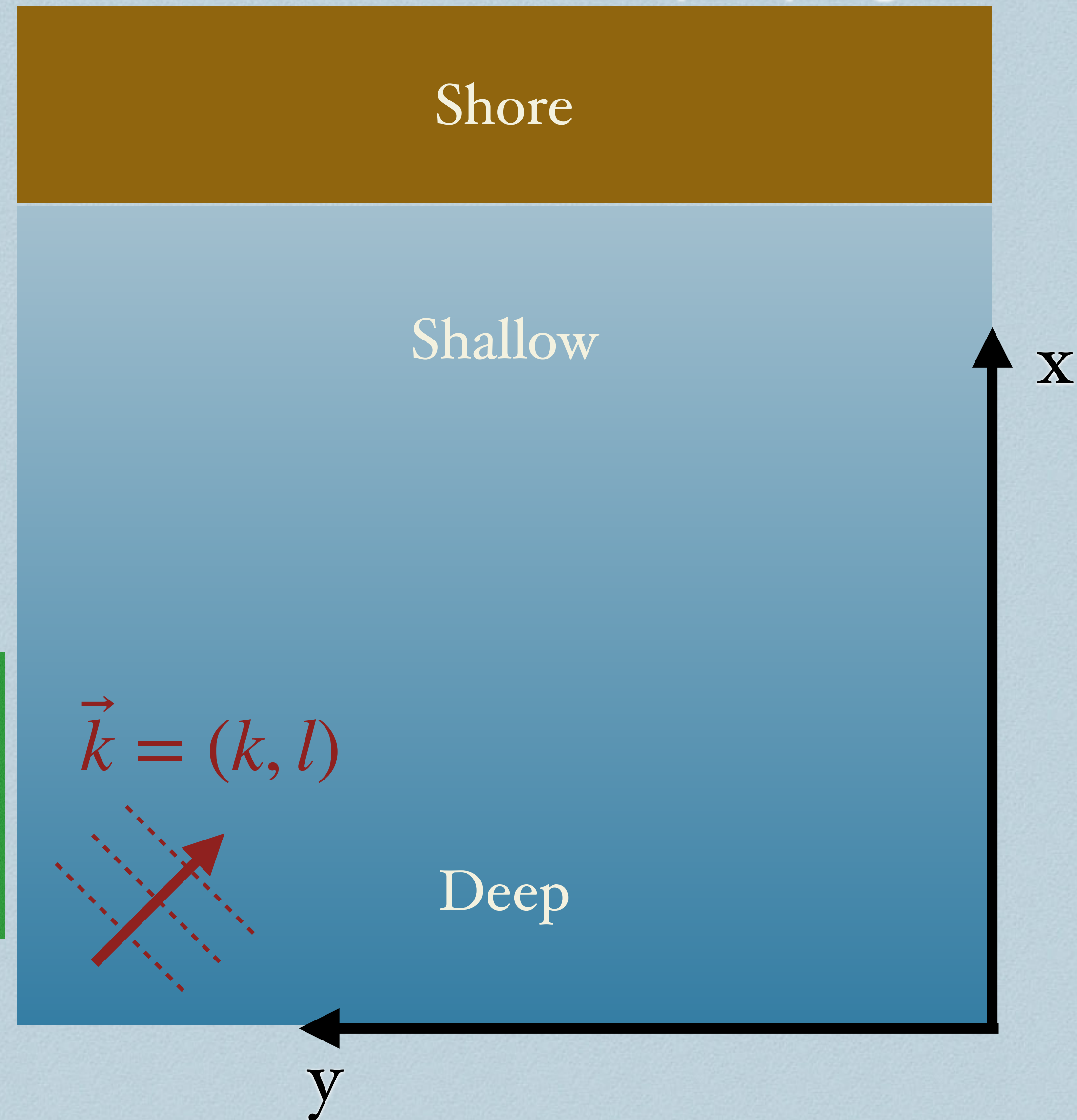
Ray Path

Change of k along the ray

Change of frequency along the ray

Change of amplitude along the ray

A 2D situation: how do waves propagate into shore?



$$\sigma = \sqrt{gH} |\vec{k}|$$

The component toward shore is k , and the component along shore is l .

The 1D case just reproduced the WKB result. But this form of the theory can handle more complex problems like the one on this slide.

- ❖ Let's consider a situation where $H(x)$ and the waves may come in at an angle: $\vec{k} = (k, l)$. So that the slowly varying solution has the form $A(x, y, t)\exp(i\theta(x, y, t))$
- ❖ The slowly varying wave parameters are $k = \partial_x \theta$, $l = \partial_y \theta$, $\sigma = -\partial_t \theta$
- ❖ The frequency again depends on a slowly varying depth: $\sigma = \Omega(k(x, t), l(x, t), H(x, y, t))$ and the group velocity is $\vec{c}_g = (c_{g_x}, c_{g_y}) = (\partial_k \sigma, \partial_l \sigma)$ and the ray path equation is a vector equation
now: $\frac{d\vec{x}}{dt} = \vec{c}_g$
- ❖ Applying the Chain rule gives 2 equations for the wave number components: $(\partial_t + c_g \partial_x)k = -\partial_H \Omega \partial_x H$, $(\partial_t + c_g \partial_y)l = -\partial_H \Omega \partial_y H$
- ❖ And a similar calculation (exercise) gives: $(\partial_t + c_g \partial_x)\sigma = \partial_H \Omega \partial_t H$
- ❖ If $H = H(x)$ these simplify considerably:

$$\sigma = \sqrt{gH(x)(k^2 + l^2)}$$

$$\vec{c}_g = (\partial_k \sigma, \partial_l \sigma) = \sqrt{gH(x)}(k, l)$$

$$\frac{d\vec{x}}{dt} = \vec{c}_g = \sqrt{gH(x)}(k, l)$$

$$(\partial_t + c_{g_x} \partial_x)k = \frac{d}{dx} \sqrt{gH(x)} k$$

$$(\partial_t + c_{g_y} \partial_y)l = 0 \implies l(x) = l_0$$

$$\begin{aligned} (\partial_t + c_{g_y} \partial_y)\sigma &= 0 \implies \sqrt{gH_0(k_0^2 + l_0^2)} = \sqrt{gH(x)(k(x)^2 + l(x)^2)} \\ &\implies k(x)^2 = \frac{H_0}{H(x)}(k_0^2 + l_0^2) - l_0^2 \end{aligned}$$

Dispersion relation and group velocity. Waves in shallow water are a bit special in this regard.

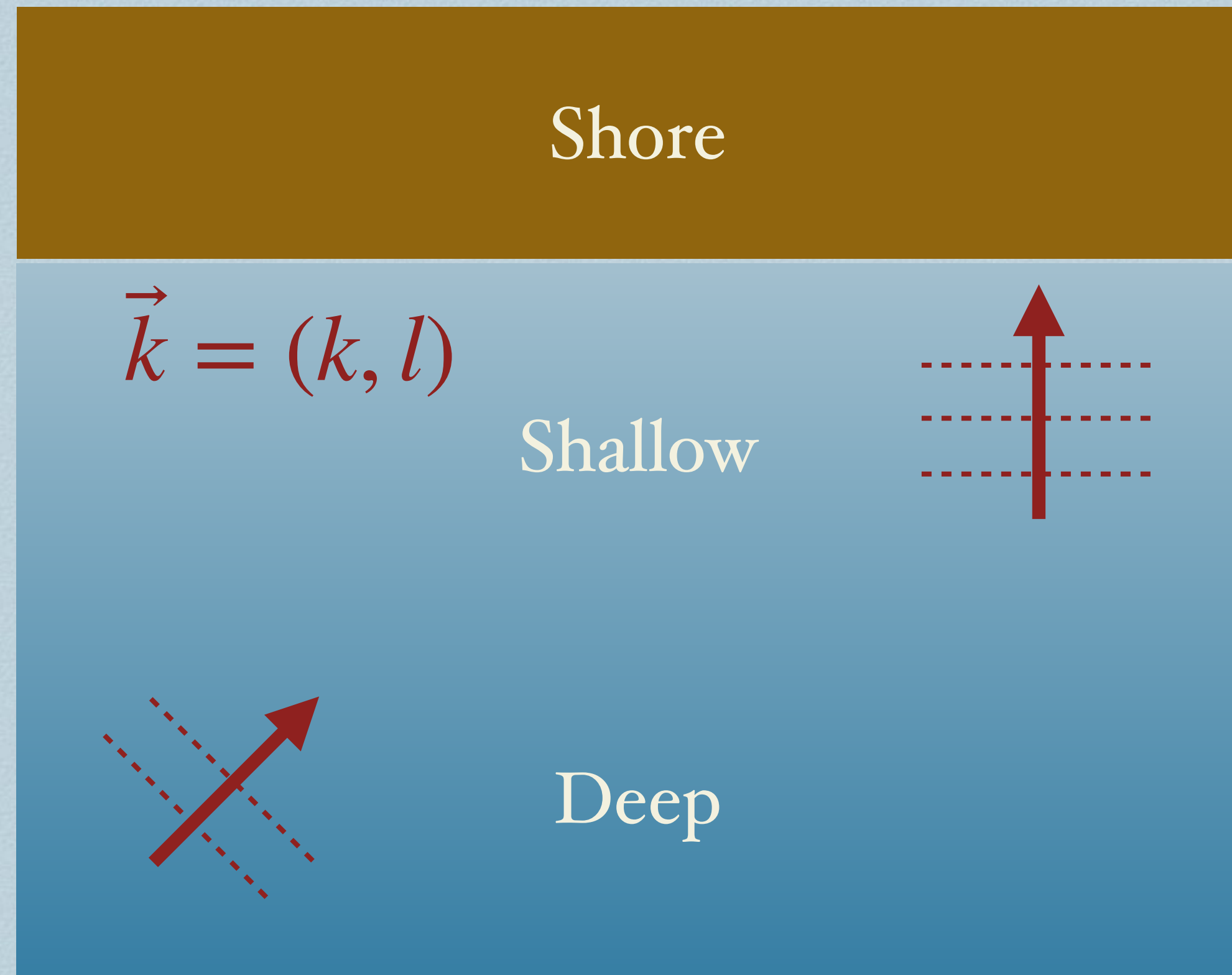
Ray Path: $H(x)$ sets the speed, but k and l set the direction

k changes along ray but l does not

As we get closer to shore we have $H(x) \rightarrow 0$ so that k grows
Since l is fixed, this means the wave number vector becomes
more and more aligned with the x -axis

$$k(x)^2 = \frac{H_0}{H(x)}(k_0^2 + l_0^2) - l_0^2$$

This would
be much
harder
to derive
with WKB
(though not
impossible)



So that waves
are observed
to move
directly
toward shore.
No matter
what
direction they
faced out in
deep water



- ❖ Of course the real world is a bit more complicated with waves breaking, like in the picture above.
- ❖ Still it is amazing how well the linear theory, and one not even solved “in full” can provide an explanation of real world observations.
- ❖ Incidentally, physicists call ray theory “Geometric Optics”.